

CURRICULUM INTELLIGENCE REPORT

HKAIVERSE Round 2 – Curriculum Intelligence Challenge

Branch: Computer Science and Engineering

College: JNTUH College of Engineering Sultanpur

Regulation: R22

Target Graduation Year: 2030

1. Introduction

I did this project to check whether the current R22 CSE curriculum is suitable for students who will graduate in 2030.

First, I checked the actual R22 CSE syllabus and the subjects taught in all eight semesters. Then I checked which subjects are still useful and which subjects need changes.

After that, I checked what has changed in the technology and software industry. I also compared the curriculum with real job descriptions from companies.

I tested an AI model with actual work from five subjects to understand what AI can do and what students still need to learn themselves.

Finally, I designed my own 2030 CSE curriculum while keeping the total degree credits at **160**.

My main idea is to “Keep the strong CSE foundation, update old parts, add important modern skills, and give students more practical experience”.

2. Current R22 CSE Curriculum

2.1 R22 Degree Structure

I studied the actual R22 B.Tech CSE curriculum followed by JNTUH College of Engineering Sultanpur. The degree contains eight semesters and 160 total credits. Each semester is planned around approximately 20 credits.

Year	Semester	Credits
I Year	I Semester	20
I Year	II Semester	20
II Year	I Semester	20
II Year	II Semester	20
III Year	I Semester	20
III Year	II Semester	20

IV Year	I Semester	20
IV Year	II Semester	20
Total		160

2.2 I Year – I Semester

Course Code	Course Name	Credits
MA101BS	Matrices and Calculus	4
CH102BS	Engineering Chemistry	4
CS103ES	Programming for Problem Solving	3
EE104ES	Basic Electrical Engineering	2
ME105ES	Computer Aided Engineering Graphics	3
CS106ES	Elements of Computer Science & Engineering	1
CH107BS	Engineering Chemistry Laboratory	1
CS108ES	Programming for Problem Solving Laboratory	1
EE109ES	Basic Electrical Engineering Laboratory	1

2.3 I Year – II Semester

Course Code	Course Name	Credits
MA201BS	Ordinary Differential Equations and Vector Calculus	4
PH202BS	Applied Physics	4
ME203ES	Engineering Workshop	2.5
EN204HS	English for Skill Enhancement	2
EC205ES	Electronic Devices and Circuits	2
CS206PC	Python Programming Laboratory	2
PH207BS	Applied Physics Laboratory	1.5
EN208HS	English Language and Communication Skills Laboratory	1
CS209ES	IT Workshop	1

2.4 II Year – I Semester

Course Code	Course Name	Credits
CS301PC	Digital Electronics	3
CS302PC	Data Structures	3
CS303PC	Computer Oriented Statistical Methods	4
CS304PC	Computer Organization and Architecture	3
CS305PC	Object Oriented Programming through Java	3
CS306PC	Data Structures Lab	1.5
CS307PC	Object Oriented Programming through Java Lab	1.5
CS308PC	Data Visualization – R Programming / Power BI	1

2.5 II Year – II Semester

Course Code	Course Name	Credits
CS401PC	Discrete Mathematics	3
SM402MS	Business Economics & Financial Analysis	3
CS403PC	Operating Systems	3
CS404PC	Database Management Systems	3
CS405PC	Software Engineering	3
CS406PC	Operating Systems Lab	1
CS407PC	DBMS Lab	1
CS408PC	Real-time Research Project / Societal Related Project	2
CS409PC	Node JS / React JS / Django	1

2.6 III Year – I Semester

Course Code	Course Name	Credits
CS501PC	Design and Analysis of Algorithms	4
CS502PC	Computer Networks	3
CS503PC	DevOps	3
PE-I	Professional Elective – I	3
PE-II	Professional Elective – II	3

CS504PC	Computer Networks Lab	1
CS505PC	DevOps Lab	1
EN508HS	Advanced English Communication Skills Lab	1
CS506PC	UI Design – Flutter	1

2.7 III Year – II Semester

Course Code	Course Name	Credits
CS601PC	Machine Learning	3
CS602PC	Formal Languages and Automata Theory	3
CS603PC	Artificial Intelligence	3
PE-III	Professional Elective – III	3
OE-I	Open Elective – I	3
CS604PC	Machine Learning Lab	1
CS605PC	Artificial Intelligence Laboratory	1
PE-III Lab	Professional Elective – III Lab	1
CS606PC	Industrial Oriented Mini Project / Internship / Skill Development Course (Big Data-Spark)	2

2.8 IV Year – I Semester

Course Code	Course Name	Credits
CS701PC	Cryptography and Network Security	3
CS702PC	Compiler Design	3
PE-IV	Professional Elective – IV	3
PE-V	Professional Elective – V	3
OE-II	Open Elective – II	3
CS703PC	Cryptography and Network Security Lab	1
CS704PC	Compiler Design Lab	1
CS705PC	Project Stage-I	3

2.9 IV Year – II Semester

Course Code	Course Name	Credits
CS801PC	Organizational Behavior	3

PE-VI	Professional Elective – VI	3
OE-III	Open Elective – III	3
Project Stage-II including Seminar	Project	11

3. Foundation vs Current Relevance

3.1 Overall Decision Table

Course Code	Subject	Semester	Decision	Short Reason
CS103ES	Programming for Problem Solving	I-I	KEEP	Fundamental programming foundation
CS301PC	Digital Electronics	II-I	KEEP	Basic computing foundation
CS302PC	Data Structures	II-I	KEEP	Important for software and interviews
CS303PC	Computer Oriented Statistical Methods	II-I	UPDATE	Important for data/ML; practical statistics can be improved
CS304PC	Computer Organization and Architecture	II-I	KEEP	Core computer understanding
CS305PC	OOP through Java	II-I	UPDATE	Keep OOP but modernize Java/tools
CS401PC	Discrete Mathematics	II-II	KEEP	Mathematical foundation
CS403PC	Operating Systems	II-II	UPDATE	Add Linux/container practical work
CS404PC	DBMS	II-II	UPDATE	Add application/database/cloud/NoSQL practice

CS405PC	Software Engineering	II-II	UPDATE	Add modern Agile/testing/CI/CD practices
CS501PC	Design and Analysis of Algorithms	III-I	KEEP	Core problem-solving skill
CS502PC	Computer Networks	III-I	UPDATE	Add modern web/cloud networking
CS503PC	DevOps	III-I	UPDATE	Connect DevOps with cloud/security/monitoring
CS601PC	Machine Learning	III-II	UPDATE	Add modern ML engineering and deployment
CS602PC	Formal Languages & Automata	III-II	KEEP	Important theoretical foundation
CS603PC	Artificial Intelligence	III-II	UPDATE	Modernize with DL, transformers, GenAI, agents
CS701PC	Cryptography & Network Security	IV-I	UPDATE	Add application/cloud/API/software security
CS702PC	Compiler Design	IV-I	KEEP	Important CS theory and language-processing foundation
CS801PC	Organizational Behavior	IV-II	KEEP	Professional/organizational skill

3.2 Detailed Subject Analysis

For my detailed subject analysis, I selected ten important subjects.

Subject	Course Code	Decision
Data Structures	CS302PC	KEEP

Object Oriented Programming through Java	CS305PC	UPDATE
Operating Systems	CS403PC	UPDATE
Database Management Systems	CS404PC	UPDATE
Design and Analysis of Algorithms	CS501PC	KEEP
Computer Networks	CS502PC	UPDATE
DevOps	CS503PC	UPDATE
Machine Learning	CS601PC	UPDATE
Artificial Intelligence	CS603PC	UPDATE
Cryptography and Network Security	CS701PC	UPDATE

4. My Subject Analysis

4.1 Data Structures – KEEP

I decided to keep Data Structures because it is one of the basic subjects of CSE.

Students still need to understand arrays, linked lists, stacks, queues, trees, graphs and other data structures.

These concepts are also useful for coding interviews and software development.

I would not remove this subject.

I would only make the practical part stronger by giving students more coding problems and real examples.

My decision: KEEP

4.2 Object Oriented Programming through Java – UPDATE

OOP is still very important in software development.

Java is also still used in many software applications.

But some older Java topics should be reduced. For example, old Applet-based development is no longer useful for modern application development.

I would keep:

- Classes and objects
- Inheritance
- Polymorphism
- Abstraction
- Encapsulation
- Exception handling
- Collections
- Multithreading

I would also introduce modern development practices such as:

- Modern Java
- Maven or Gradle
- JUnit testing
- Git
- JSON
- REST API basics
- Debugging

My decision: UPDATE

4.3 Operating Systems – UPDATE

Operating Systems is still an important CSE subject.

Students should understand:

- Processes
- Threads
- CPU scheduling
- Memory management
- File systems
- Deadlocks
- Synchronization

I would keep these concepts.

But students should also get practical exposure to:

- Linux
- Process and thread commands
- Virtual machines
- Containers
- Basic Docker usage

This will help students connect OS concepts with modern software environments.

My decision: UPDATE

4.4 Database Management Systems – UPDATE

Database knowledge is still very important because almost every software application stores data.

I would keep:

- SQL
- Relational databases
- Normalization
- Transactions

- Indexing
- Queries
- Constraints

But I would add more practical work related to:

- REST APIs
- Database connections
- Query optimization
- Database security
- Cloud databases
- Basic NoSQL concepts

Students should not only write SQL queries. They should also learn how a database is connected to an actual application.

My decision: UPDATE

4.5 Design and Analysis of Algorithms – KEEP

Algorithms are still very important for software development and technical interviews.

Students should continue learning:

- Searching
- Sorting
- Greedy algorithms
- Dynamic programming
- Graph algorithms
- Time complexity
- Space complexity

I would keep the subject.

But I would connect the theory with more coding practice and real problems.

My decision: KEEP

4.6 Computer Networks – UPDATE

Networking basics are still necessary.

Students should understand:

- OSI model
- TCP/IP
- IP addressing
- Routing
- TCP
- UDP
- DNS

- HTTP
- Network security basics

I would also introduce modern topics such as:

- IPv6
- HTTPS and TLS
- HTTP/2 and HTTP/3
- Cloud networking
- Virtual networks
- Basic network monitoring

My decision: UPDATE

4.7 DevOps – UPDATE

DevOps is useful because software companies need to build, test and deploy applications.

The R22 syllabus already gives some practical exposure to tools such as:

- Git
- Jenkins
- Docker
- Kubernetes
- Selenium

I would keep the main idea of DevOps.

But I would connect it more with:

- Cloud
- CI/CD
- Infrastructure basics
- Monitoring
- Security
- Containers
- Kubernetes
- DevSecOps

The important point is that students should understand the complete process instead of learning tools separately.

My decision: UPDATE

4.8 Machine Learning – UPDATE

Machine Learning is important for modern CSE students.

The existing topics such as:

- Regression

- Classification
- Clustering
- Decision trees
- SVM
- Neural networks

are still useful.

But students also need practical knowledge of:

- Python
- NumPy
- Pandas
- Scikit-learn
- Data preprocessing
- Feature selection
- Model evaluation
- Experiment tracking
- Model deployment

Therefore, I would keep the basic algorithms but update the practical part.

My decision: UPDATE

4.9 Artificial Intelligence – UPDATE

AI is already present as a core subject in R22.

Therefore, I do not think the solution is simply to say “**add AI.**”

AI itself has changed.

Students should learn about:

- Machine Learning
- Deep Learning
- Neural networks
- Transformers
- Embeddings
- Generative AI
- Large Language Models
- Prompt Engineering
- AI agents
- AI evaluation
- Responsible AI

The aim is to connect traditional AI knowledge with modern AI systems.

My decision: UPDATE

4.10 Cryptography and Network Security – UPDATE

Security fundamentals are still important.

Students should continue learning:

- Encryption
- Decryption
- RSA
- AES
- Hash functions
- Digital signatures
- Authentication

But modern applications also need:

- API security
- Identity and access management
- Cloud security
- Secure coding
- Software supply-chain security
- Threat modelling
- Secure software development

Therefore, I would update the subject instead of removing it.

My decision: UPDATE

5. What Changed in the Real World?

I checked current technology and industry sources to understand what has changed after the R22 curriculum was designed.

I found eight important changes.

5.1 Generative AI became much more common

Generative AI is now used for text, code, images and many other tasks.

This affects:

- Machine Learning
- Artificial Intelligence

So students should understand how GenAI systems work and how to use and check them.

5.2 AI-assisted software development became common

Developers are increasingly using AI tools while writing and debugging code.

This affects:

- Java
- Algorithms

- DevOps
- Machine Learning
- AI

Students should learn how to use AI tools but also verify the code instead of blindly accepting the answer.

5.3 AI agents became an important area

AI systems are becoming capable of using tools and completing multiple steps.

This affects:

- Artificial Intelligence
- DevOps
- Software development

Students should understand the basic idea of AI agents and how to evaluate their output.

5.4 Cloud-native and Kubernetes became important

Modern applications are often deployed using containers and cloud platforms.

Kubernetes is also widely used for managing container-based applications.

This affects:

- DevOps
- Cloud
- Distributed Systems

Therefore, I would connect DevOps with cloud and practical deployment.

5.5 Java continues to change

Modern Java releases continue to add new features.

This affects:

- Object Oriented Programming through Java

Therefore, Java education should use current Java development practices instead of spending too much time on older technologies.

5.6 Software security became more important

Software supply-chain problems and application security are important concerns.

This affects:

- Cyber Security
- Cryptography and Network Security
- DevOps

Students should learn secure development and basic software security practices.

5.7 Responsible AI became important

AI systems can give incorrect, biased or unsafe results.

Students therefore need to understand:

- AI evaluation
- Risks
- Responsible use
- Checking AI output

This affects:

- Machine Learning
- Artificial Intelligence

5.8 Python became more important for AI and data work

Python is widely used for:

- Machine Learning
- Data analysis
- AI
- Automation

Therefore, Python should be used more in practical AI and ML work.

6. Curriculum vs Industry

I compared my curriculum with three real job descriptions.

The three jobs were:

1. Amazon – 2026 Graduate Software Development Engineer
2. Amgen – Associate Software Engineer – Full Stack
3. WRI India – Program Associate – Software Engineer

6.1 Amazon – Graduate Software Development Engineer

The job expects skills such as:

- Java/Python or another programming language
- Data Structures
- Algorithms
- OOP
- SQL/NoSQL
- Cloud
- Git
- CI/CD
- Debugging
- AI tools
- Cloud-native development

What R22 already provides

R22 has:

- Data Structures
- Java/OOP
- DBMS
- Algorithms
- DevOps
- AI
- ML

Gap I noticed:

The main gap is that these skills are taught separately.

Students need more practical experience connecting:

coding → Git → testing → deployment → cloud → monitoring

My conclusion: The foundation is good, but practical software development should be stronger.

6.2 Amgen – Associate Software Engineer – Full Stack

The Amgen job expects skills such as:

- React
- JavaScript/TypeScript
- Python
- SQL
- REST APIs
- Cloud
- Docker
- Kubernetes
- Testing
- DevOps
- AI/GenAI
- Security
- Monitoring

What R22 already provides

R22 has related subjects such as:

- DBMS
- Software Engineering
- Node JS/React JS/Django
- DevOps
- ML
- AI
- Security

Gap I noticed:

The subjects are present, but students may not get enough experience building one complete application using all these skills.

My conclusion: Students should get more end-to-end project work.

6.3 WRI India – Program Associate – Software Engineer

The WRI India software role expects:

- React/Vue
- Python/Node.js
- PostgreSQL
- REST APIs
- Git/GitHub
- Docker
- Cloud
- CI/CD
- AI/ML
- LLM APIs

What R22 already provides

R22 has:

- DBMS
- Web development
- DevOps
- ML
- AI
- Software Engineering

Gap I noticed:

There is less direct focus on:

- LLM APIs
- Cloud deployment
- Practical full-stack applications
- Complete project deployment

My conclusion: The curriculum should connect existing subjects with current development practices.

7. My AI Experiment

I tested GPT-5.6 Luna with actual coursework from five subjects.

Test date: 14 September 2026

The five subjects were:

1. Cryptography and Network Security
2. Machine Learning
3. Compiler Design
4. Java

5. DBMS

7.1 CNS

Prompt used: Solve RSA for $p = 7$, $q = 11$, $e = 17$ and message $m = 8$. Show all steps.

What AI gave

The model calculated:

- $n = 77$
- $\phi(n) = 60$
- $d = 53$
- Public key = $(17, 77)$
- Private key = $(53, 77)$
- Ciphertext = 57
- Decrypted message = 8

My result: Pass for this numerical problem.

The answer was useful because it showed the calculation steps.

7.2 Machine Learning

Prompt used: Explain the K-Means clustering algorithm.

What AI gave

AI explained:

- What clustering means
- Selecting K
- Assigning data points
- Finding centroids
- Repeating the process
- Stopping when the clusters become stable

My result: Pass for theory.

But the student still needs to understand how to select K and how to interpret the final clusters.

7.3 Compiler Design

Prompt used: Write a LEX program to identify keywords, identifiers, numbers, operators and special symbols from a given input program.

What AI can do

AI can generate:

- LEX code
- Regular expressions
- Explanation
- Sample input/output

My result: Requires testing.

I cannot say that the generated program will definitely pass the college lab because it needs to be compiled and executed in the actual environment.

7.4 Java

Prompt used: Write a Java program for a simple calculator GUI as required in the Java lab.

What AI can do

AI can generate:

- GUI design
- Buttons
- Text fields
- Arithmetic operations
- Java code

My result: Requires testing.

The code needs to be compiled and executed to check whether it works correctly.

7.5 DBMS

Prompt used: Explain and demonstrate the creation of INSERT, DELETE and UPDATE triggers in SQL with suitable examples.

What AI can do

AI can provide:

- Trigger syntax
- SQL examples
- Explanation
- Sample tables

My result : Requires testing.

SQL trigger syntax can depend on the database system being used.

7.6 My AI Experiment Conclusion

From this experiment, I found that AI can:

- Explain concepts
- Solve some numerical problems
- Generate code
- Generate SQL
- Generate LEX programs
- Give examples quickly

But students still need to:

- Understand the question
- Check the answer

- Run the code
- Find errors
- Debug
- Test the output
- Decide whether the solution is actually correct

So I do not think AI makes students unnecessary.

Instead: AI makes understanding, checking, testing and debugging even more important.

8. My 2030 Curriculum

I designed my own 2030 curriculum based on the evidence I collected.

The most important rule was: I should not simply keep adding subjects because the degree already has 160 credits.

Therefore, my final curriculum has 160 total credits and 20 credits in every semester.

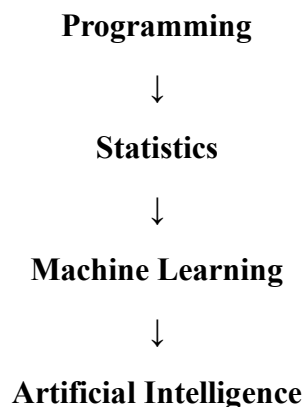
8.1 My Main Changes

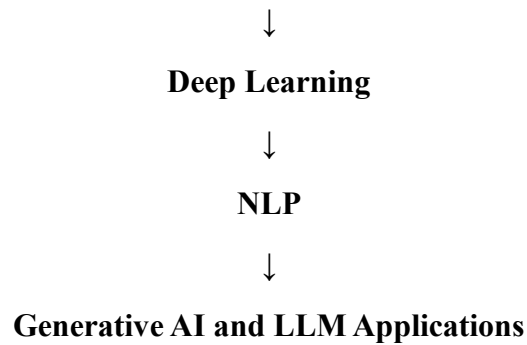
I strengthened the following areas:

- Deep Learning
- Deep Learning Lab
- Prompt Engineering
- AI-assisted development
- NLP
- Cyber Security
- Generative AI
- LLM applications
- Cloud-native development
- AI/ML practical engineering
- Testing
- Secure software development

9. My 2030 AI Learning Path

My proposed AI learning path is:

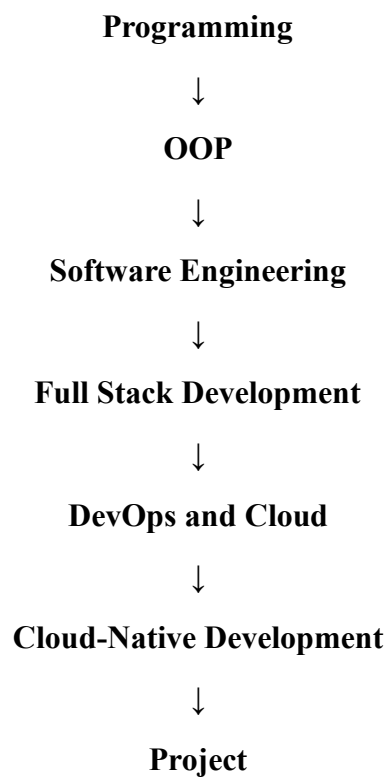




This gives students a step-by-step path instead of suddenly introducing advanced AI topics.

10. My 2030 Software Development Path

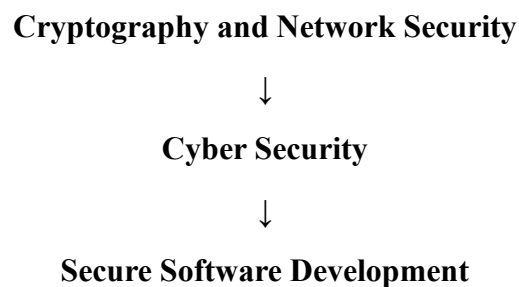
My proposed software path is:



The goal is that a student should be able to build, test and deploy a complete application.

11. My 2030 Security Path

My proposed security path is:





DevSecOps

This connects security with actual software development.

12. R22 vs R25 vs My 2030 Curriculum

I checked the R25 changes because they show the direction in which the curriculum is already moving.

Area	R22	R25 Constituent College	My 2030
Machine Learning	III-II	II-II	Kept with proper prerequisite path
Deep Learning	PE-VI option	Core III-II + Lab	Core III-II + Lab
Prompt Engineering	Not present	1 credit	1 credit
Generative AI	Not present	PE option	Dedicated 3-credit subject
NLP	PE-II	Core IV-I	Strengthened modern NLP
Cyber Security	PE-IV	Core IV-I + Lab	Core + practical lab
DevOps	3 + 1 credits	1-credit skill course in constituent college	3-credit DevOps & Cloud
Statistics	II-I	II-II	Kept before ML in learning path

13. Why I Did Not Simply Copy R25

R25 already makes many useful changes.

For example:

- Machine Learning is moved earlier.
- Deep Learning becomes core.
- Prompt Engineering is added.
- NLP becomes core.
- Cyber Security becomes core.

I used these changes as evidence.

But I did not simply copy R25.

My 2030 curriculum tries to connect: AI + Software + Cloud + Security + Testing + Projects

For example, I kept a stronger DevOps and Cloud path instead of reducing DevOps too much.

I also kept Statistics before Machine Learning because I think the learning order is important.

I gave Generative AI and LLM applications more dedicated space because they are becoming important in current software development.

14. Credit Balance

My final curriculum has: **8 semesters × 20 credits = 160 credits**

I did not increase the total degree credits.

The new areas are funded by changing and restructuring existing curriculum space.

For example:

- Deep Learning is given core space by restructuring AI/ML content.
- Deep Learning Lab uses practical credit space.
- Prompt Engineering is kept as a small 1-credit skill.
- Cyber Security gets stronger core and practical coverage.
- GenAI and LLM Applications get dedicated space by reducing repeated or older AI emphasis.
- AI/ML Engineering Practice uses a small practical skill-development space.
- Cloud-native topics are connected with the existing DevOps and systems areas.

15. What Would Make Me Wrong?

I do not want to make my curriculum changes only because I think they are good.

There are some things that could prove me wrong.

1. Current jobs may not require these skills

If most graduate jobs do not actually require the modern skills I identified, then some additions may not be necessary.

2. Employers may say R22 is already enough

If employers say that the existing curriculum prepares students well, then I would need to reduce some changes.

3. The new curriculum may be too difficult to teach

If the proposed subjects create too much workload for students or teachers, the curriculum would need to be simplified.

4. R25 may already solve the main problems

If R25 is found to be better than my proposed design, I would need to reconsider my changes.

5. AI may not be as useful as expected

If real testing shows that AI-generated solutions are often incorrect, students will need even stronger verification and debugging skills.

16. One Thing I Changed My Mind About

At first, I thought the main problem was: **Students need more AI subjects.**

After checking the R25 curriculum and current industry evidence, I changed my opinion.

AI is already being added to the curriculum.

So I now think the bigger problem is: **Students need strong CSE foundations plus modern AI, software development, cloud, security and practical skills.**

Adding many AI subjects alone will not solve the problem.

17. Interview Status

The HKAIVERSE challenge requires a 20-minute conversation with a CSE faculty member or a recent graduate working in industry.

I tried to arrange this conversation, but I was not able to complete it.

I do not want to create a fake interview or write answers that somebody did not actually give.

Therefore, I have marked the interview part as incomplete.

I have not used an imaginary interview as evidence for my curriculum decisions.

My other findings are based on:

- Actual R22 syllabus
- R25 information
- Current industry sources
- Real job descriptions
- Actual coursework
- AI experiment

18. Final Findings

After doing this research, I found that the R22 curriculum has a strong CSE foundation.

Subjects such as:

- Data Structures
- Algorithms
- Operating Systems
- DBMS
- Networks
- Programming

are still useful.

The main problem is that some subjects need updates and students need more practical experience.

The important modern areas are:

- AI
- Generative AI
- LLMs
- Cloud

- DevOps
- Cyber Security
- Testing
- APIs
- Git
- Deployment
- AI-assisted development

The subjects should not be treated as completely separate.

Students should get experience connecting them.

19. Final Conclusion

My biggest finding is: The R22 CSE curriculum has strong foundations, but several subjects need targeted updates so that students graduating in 2030 can connect those foundations with modern AI, software development, cloud, security and practical engineering work.

I do not think all old subjects should be removed.

Basic CSE concepts are still important.

My approach is:

Keep the foundation → Update old parts → Reduce unnecessary content → Add important modern skills → Connect everything through practical projects.

The first major change I would make is to create a connected learning path where students move from programming and core CSE subjects to AI/ML, software development, cloud, security and practical project work.

This would make the curriculum more useful for students graduating in 2030.

20. Main Sources Used

The detailed source information is recorded in the separate Sources_Log.xlsx file.

The main sources I used include:

- JNTUH R22 CSE syllabus
- HKAIVERSE Curriculum Intelligence Challenge document
- Stanford AI Index
- Stack Overflow Developer Survey
- CNCF Cloud Native Survey
- Oracle Java documentation
- OWASP Top 10
- NIST Generative AI Profile
- Amazon job description
- Amgen job description

- WRI India job description

Appendix A – R22 CSE Course Structure

I Year – I Semester

Course Code	Course Name	Credits
MA101BS	Matrices and Calculus	4
CH102BS	Engineering Chemistry	4
CS103ES	Programming for Problem Solving	3
EE104ES	Basic Electrical Engineering	2
ME105ES	Computer Aided Engineering Graphics	3
CS106ES	Elements of Computer Science & Engineering	1
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I Year – II Semester

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CS206PC	Python Programming Laboratory	2
PH207BS	Applied Physics Laboratory	1.5
EN208HS	English Language and Communication Skills Laboratory	1

Course Code	Course Name	Credits
CS209ES	IT Workshop	1

II Year – I Semester

Course Code	Course Name	Credits
CS301PC	Digital Electronics	3
CS302PC	Data Structures	3
CS303PC	Computer Oriented Statistical Methods	4
CS304PC	Computer Organization and Architecture	3
CS305PC	Object Oriented Programming through Java	3
CS306PC	Data Structures Lab	1.5
CS307PC	Object Oriented Programming through Java Lab	1.5
CS308PC	Data Visualization – R Programming / Power BI	1

II Year – II Semester

Course Code	Course Name	Credits
CS401PC	Discrete Mathematics	3
SM402MS	Business Economics & Financial Analysis	3
CS403PC	Operating Systems	3
CS404PC	Database Management Systems	3
CS405PC	Software Engineering	3
CS406PC	Operating Systems Lab	1
CS407PC	DBMS Lab	1
CS408PC	Real-time Research Project / Societal Related Project	2
CS409PC	Node JS / React JS / Django	1

III Year – I Semester

Course Code	Course Name	Credits
CS501PC	Design and Analysis of Algorithms	4
CS502PC	Computer Networks	3
CS503PC	DevOps	3
PE-I	Professional Elective – I	3
PE-II	Professional Elective – II	3
CS504PC	Computer Networks Lab	1
CS505PC	DevOps Lab	1
EN508HS	Advanced English Communication Skills Lab	1
CS506PC	UI Design – Flutter	1

III Year – II Semester

Course Code	Course Name	Credits
CS601PC	Machine Learning	3
CS602PC	Formal Languages and Automata Theory	3
CS603PC	Artificial Intelligence	3
PE-III	Professional Elective – III	3
OE-I	Open Elective – I	3
CS604PC	Machine Learning Lab	1
CS605PC	Artificial Intelligence Laboratory	1
PE-III Lab	Professional Elective – III Lab	1
CS606PC	Industrial Oriented Mini Project / Internship / Skill Development Course (Big Data-Spark)	2

IV Year – I Semester

Course Code	Course Name	Credits
CS701PC	Cryptography and Network Security	3
CS702PC	Compiler Design	3

Course Code	Course Name	Credits
PE-IV	Professional Elective – IV	3
PE-V	Professional Elective – V	3
OE-II	Open Elective – II	3
CS703PC	Cryptography and Network Security Lab	1
CS704PC	Compiler Design Lab	1
CS705PC	Project Stage-I	3

IV Year – II Semester

Course Code	Course Name	Credits
CS801PC	Organizational Behavior	3
PE-VI	Professional Elective – VI	3
OE-III	Open Elective – III	3
Project Stage-II including Seminar Project		11

Total R22 Credits: 160

R22 Professional Electives

PE-I

- CS511PE – Quantum Computing
- CS512PE – Advanced Computer Architecture
- CS513PE – Data Analytics
- CS514PE – Image Processing
- CS515PE – Principles of Programming Languages

PE-II

- CS521PE – Computer Graphics
- CS522PE – Embedded Systems
- CS523PE – Information Retrieval Systems
- CS524PE – Distributed Databases
- CS525PE – Natural Language Processing

PE-III

- CS631PE – Full Stack Development
- CS632PE – Internet of Things
- CS633PE – Scripting Languages
- CS634PE – Mobile Application Development
- CS635PE – Software Testing Methodologies

PE-IV

- CS741PE – Graph Theory
- CS742PE – Cyber Security
- CS743PE – Soft Computing
- CS744PE – Cloud Computing
- CS745PE – Ad hoc & Sensor Networks

PE-V

- CS751PE – Advanced Algorithms
- CS752PE – Agile Methodology
- CS753PE – Robotic Process Automation
- CS754PE – Blockchain Technology
- CS755PE – Software Process & Project Management

PE-VI

- CS861PE – Computational Complexity
- CS862PE – Distributed Systems
- CS863PE – Deep Learning
- CS864PE – Human Computer Interaction
- CS865PE – Cyber Forensics

R22 Open Electives**OE-I**

- CS611OE – Data Structures
- CS612OE – Database Management Systems

OE-II

- CS721OE – Operating Systems

- CS722OE – Software Engineering

OE-III

- CS831OE – Algorithms Design and Analysis
- CS832OE – Introduction to Computer Networks

Why I Included This Appendix

I included the complete R22 course structure to show what I actually checked before making my 2030 curriculum proposal. I used the complete eight-semester structure, credits and elective choices as the starting point. My proposed changes are based on this R22 structure and not on a generic CSE curriculum.

Appendix B – 2030 Credit Funding and Balance

My 2030 curriculum should not increase the total degree credits. The R22 degree has 160 credits, so I kept the total at 160 credits.

I added some new topics such as Deep Learning, Prompt Engineering, Cyber Security, Generative AI and Cloud-Native development. To keep the total credits at 160, I used existing credit space and reduced repeated or older content instead of simply adding more subjects.

Why I did not increase the total credits

I did not want students to have a very large syllabus just because technology is changing quickly. New topics should replace or improve older/repeated content where possible.

For example, instead of teaching old technologies for a long time and then adding new AI topics on top of them, I would reduce outdated material and use that learning time for modern tools and practical work.

Final Credit Balance

My proposed 2030 curriculum has:

- I Year I Semester – 20 credits
- I Year II Semester – 20 credits
- II Year I Semester – 20 credits
- II Year II Semester – 20 credits
- III Year I Semester – 20 credits
- III Year II Semester – 20 credits
- IV Year I Semester – 20 credits

- IV Year II Semester – 20 credits

Total = 160 credits

Therefore:

R22 total = 160 credits

My 2030 total = 160 credits

Difference = 0 credits

My conclusion

The main idea is not to make the degree longer. My idea is to keep the important CSE foundation, remove or reduce outdated and repeated parts, and use that space for current skills.

This keeps the degree manageable for students while making it more suitable for students graduating in 2030.

Final Validation Addendum

I added this final check section after reviewing my complete report. It makes some important evidence easier to find. I am not changing my main findings. I am only making the links between my evidence, decisions and 2030 proposal clearer.

1. Redesign Credit Calculation

The R22 degree has 160 total credits. 20% of 160 = 32 credits. 25% of 160 = 40 credits. So, the expected redesign range is 32 to 40 credits.

My 2030 proposal changes and restructures about 36 credits of learning space. $36 / 160 \times 100 = 22.5\%$. Therefore, my redesign stays within the 20–25% range.

2. Credit Funding for New and Strengthened Areas

I did not add extra credits to the degree. The total remains 160 credits. The table below shows where the changed learning space comes from. These are curriculum changes and restructuring, not an increase in the degree size.

2030 area	Credits changed	Funded by restructuring	My reason
Deep Learning + Lab	5	AI/ML elective and practical space	Deep Learning is moved from optional study to stronger structured learning.
Prompt Engineering and AI-assisted Development	2	Reduced repeated skill/theory space	Students need to use AI tools and also verify the output.
Generative AI and LLM Applications	3	Restructured AI content and elective space	GenAI and LLM use were not directly covered in R22.
AI/ML Engineering Practice	3	Practical/project space	Students need deployment,

			evaluation and testing practice.
Cyber Security practical strengthening	4	Security elective/core restructuring	Security needs stronger practical work.
Cloud-Native Development	4	DevOps and systems restructuring	Cloud, containers and deployment should be connected.
Testing and Secure Software Development	3	Software Engineering restructuring	Testing and secure development should be practiced together.
Modern Java, tools and version control updates	4	Reduced outdated Java coverage	OOP stays, but older parts can be reduced for current tools.
NLP strengthening	4	AI elective/core restructuring	NLP is important for modern AI applications.
Other practical integrations	4	Rearranged lab/project activities	Git, APIs, monitoring and deployment are connected through practice.

3. Prerequisite Summary

Subject / Area	Main prerequisite
Machine Learning	Programming and Statistics
Artificial Intelligence	Data Structures and Algorithms
Deep Learning	Machine Learning
NLP	Machine Learning and AI basics
Generative AI and LLM Applications	AI, Deep Learning and NLP basics
Full Stack Development	OOP, DBMS and Software Engineering
DevOps and Cloud	Operating Systems, Networks and Software Engineering
Cloud-Native Development	DevOps and Cloud
Cyber Security	Computer Networks and Cryptography basics
Secure Software Development	Software Engineering and Cyber Security

I used these prerequisites so that advanced subjects are not introduced suddenly. For example, Deep Learning comes after Machine Learning, and Generative AI comes after students already understand AI and related foundations.

4. What I Could Not Verify

- The original lab grading rubrics used for every subject.
- The exact marks given for old submitted lab programs.
- The exact software setup used by the faculty for every lab.
- Direct employer feedback about the complete R22 curriculum.
- A completed 20-minute faculty or industry interview.
- Whether every AI-generated Java, LEX and DBMS solution would pass in the original college environment.

5. AI Claim I Found Weak

One claim I found weak was the idea that an AI-generated answer can automatically be treated as a correct or passing solution.

In my experiment, GPT-5.6 Luna gave useful answers for CNS, Machine Learning, Compiler Design, Java and DBMS. However, the Java, LEX and DBMS outputs still required compiling, running or testing in the correct environment. Because of this, I did not treat AI output as automatically correct.

My final view is: AI-generated output is a useful starting point, but the student still has to understand, test, check and debug it.

6. Five-Subject AI Experiment at a Glance

Subject	Actual coursework used	Result	What still needed checking
CNS	RSA numerical assignment	Pass for the checked numerical problem	Verify calculations
Machine Learning	K-Means assignment/theory	Pass for explanation	Choose K and interpret results
Compiler Design	LEX program assignment	Requires testing	Compile and run
Java	Calculator GUI lab question	Requires testing	Compile, execute and debug
DBMS	Trigger questions	Requires testing	Check syntax in the actual DBMS

7. Source and Date Note

For important current claims, I used the source information recorded in Sources_Log.xlsx. The main sources include the JNTUH R22 syllabus, Stanford AI Index, Stack Overflow Developer Survey, CNCF Cloud Native Survey, Oracle Java documentation, OWASP Top 10, NIST Generative AI Profile, and the three job descriptions from Amazon, Amgen and WRI India. The publication year or access/check date is recorded in the source log.

I did not invent interview evidence. The interview remains honestly marked as incomplete in this report.